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| Michael Galileo Hasian (201344020) | **Rangkuman Minggu ke-12** | Senin, 21 November 2022 |
| | | Aplikasi Mikrokontroler |
| 3-TNK | | Ferry Satria, BSEE, M.T.<br>Rahmawati Hasanah, S.ST.,M.T. |

ITU - T ITU

(The International Telecomunication ) → badan dari United Nations

ITU – T Y.2060 (tentang IoT)

Frameworks and functional architecture models

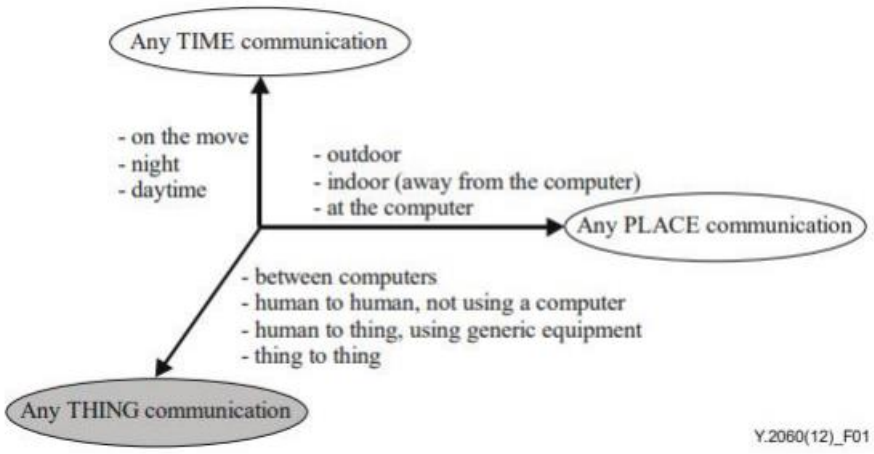
Devices : memiliki kemampuan untuk dapat berkomunikasi, bisa membaca data sensor, mengolah, menyimpan, menampilkan

IoT : A global infrastructure for the information society, enabling advanced services by interconnecting (physical and virtual) things based on existing and evolving interoperable information and communication technologies.

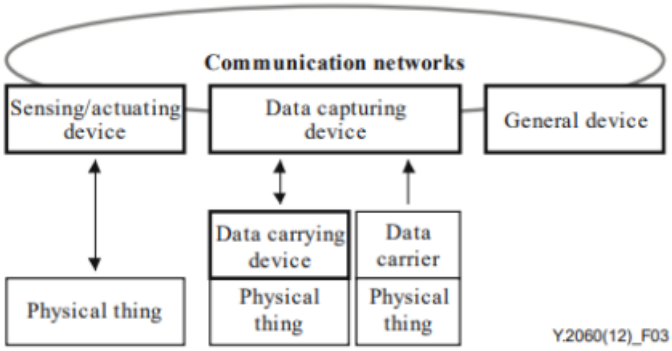
Interkoneksi : fisik dan virtual

*noted : keamanan dan privasi

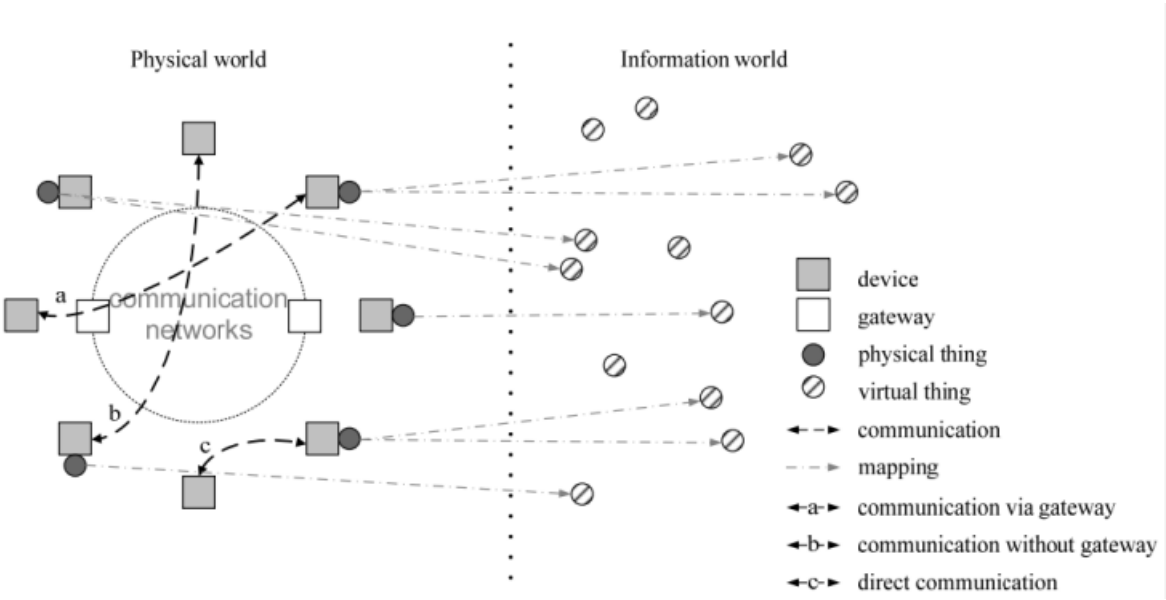
Data capture, memproses data dan mengirimkannya




Physical thing : represented in the information world via one or more virtual things (mapping), but a virtual thing can also exist without any associated physical thing.




Virtual word :




Direct communication : tanpa perlu adanya jaringan untuk berkomunikasi

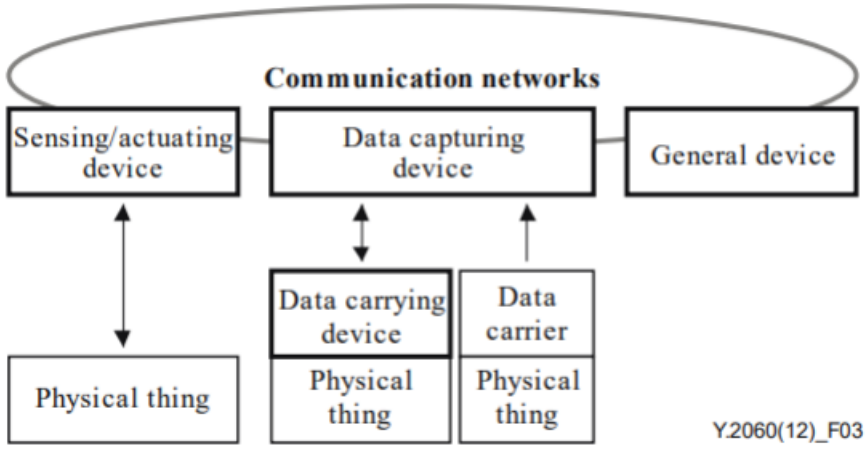
Karakteristik fundamental

- Interconnectivity: With regard to the IoT, anything can be interconnected with the global information and communication infrastructure.
- Things-related services: The IoT is capable of providing thing-related services within the constraints of things, such as privacy protection and semantic consistency between physical things and their associated virtual things. In order to provide thing-related services within the constraints of things, both the technologies in physical world and information world will change.
- Heterogeneity: The devices in the IoT are heterogeneous as based on different hardware platforms and networks. They can interact with other devices or service platforms through different networks.
- Dynamic changes: The state of devices change dynamically, e.g., sleeping and waking up, connected and/or disconnected as well as the context of devices including location and speed. Moreover, the number of devices can change dynamically.

- Enormous scale: The number of devices that need to be managed and that communicate with each other will be at least an order of magnitude larger than the devices connected to the current Internet. The ratio of communication triggered by devices as compared to communication triggered by humans will noticeably shift towards device-triggered communication. Even more critical will be the management of the data generated and their interpretation for application purposes. This relates to semantics of data, as well as efficient data handling.

Model Referensi IoT (4 Layer) :
- Application layer
- Service support and application support layer
- Network layer
- Device layer.

Device : Arduino/system microkontroller
Physical thing : sensor/actuator
Virtual thing : temperature disekitar sensor temperature
Communication network : jaringan internet, jaringan private LAN, Bluetooth
Gateway : yg menghubungkan perangkat komunikasi dengan jaringan komunikasi ➔ wifi router




Capture → simpan → olah → dikirimkan

Minimum requirement IoT:
- Data carrying
- Data capturing
- Sensoring & actuating
- Fundamentals Characteristics :

- Interconnectivity
- Things related services
- Heterogeneity
- Synamic changes
- Enormous scale

High level requirement :

IoT reference model :

1. Application layer
2. service support and application support layer
3. network layer
4. device layer

